

Clinical guideline foundation

TiTrATE · Barany ICVD · AHA/ASA · Framingham · NHANES

L1

Combinatorial
enumeration

L2

Bayesian
network

L3

Rule-based
expert system

L4

XAI-by-design
provenance

L5

Counterfactual
reasoning

Weighted ensemble of Layers 1-3

$$D_{\text{final}} = w_1 C + w_2 B + w_3 R$$

$w^* = [0.10, 0.10, 0.80]$, grid-search selected; Layers 4-5 act downstream

Four-dimensional validation

statistical: KL / JS / Wasserstein diagnostic: ROC-AUC, sensitivity, specificity

clinical: TiTrATE constraint satisfaction explainability: guideline traceability

SynDX-Hybrid synthetic cohort

N = 10,000 records · 150 features per layer · 7-class common taxonomy

100% guideline-traceable · 72.4% counterfactual consistency